

CSE200 • B2 • Group 2

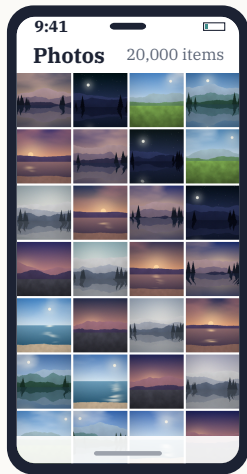
VARIATIONAL AUTOENCODER

Presented by

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Department of CSE, BUET

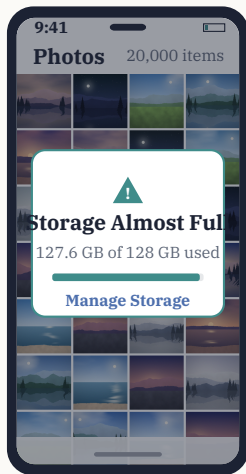
The Problem: Storage Almost Full



Imagine your phone has

20,000 photos

The Problem: Storage Almost Full



Storage Almost **Full!**

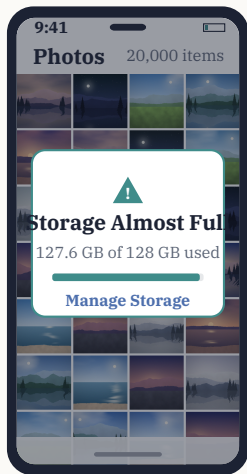
Storage

127.6 GB of 128 GB used



■ Photos 112 GB ■ Apps 10 GB ■ System 5.6 GB

The Problem: Storage Almost Full



Storage Almost **Full!**

Storage 127.6 GB used **only 0.4 GB free**



■ Photos 112 GB ■ Apps 10 GB ■ System 5.6 GB

90% Storage
Occupied by Photos!

What do you do?

What do you do?



Delete Photos

not ideal: once gone, they're gone forever

What do you do?



Delete Photos

not ideal: once gone, they're gone forever



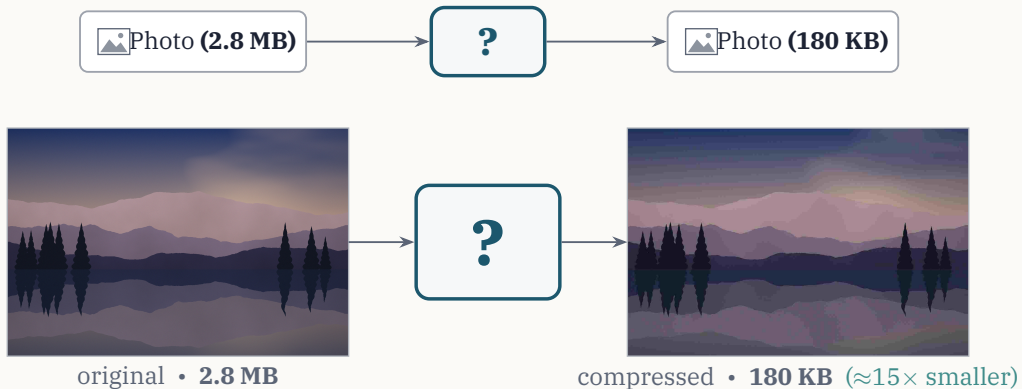
A Smarter Solution

keep every photo, but use lesser space

Compression Saves Storage



Compression Saves Storage



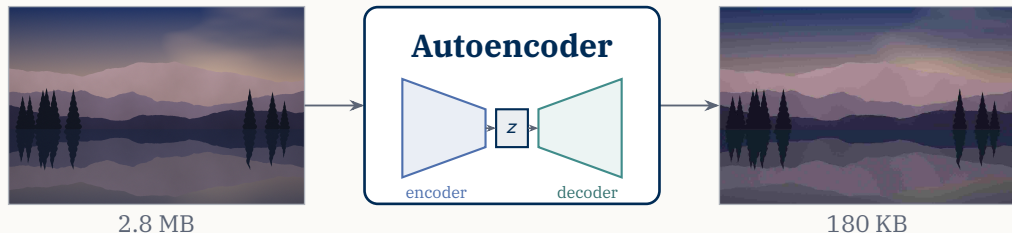
Compressed image is **15 times smaller**

Enter the Autoencoder



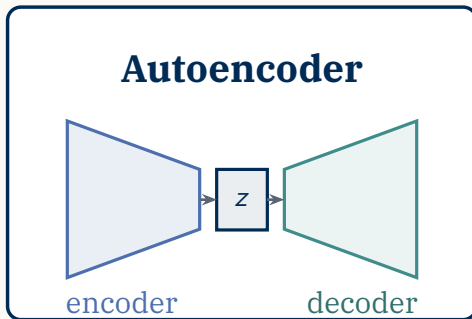
What goes inside the **mysterious middle box**?

Enter the Autoencoder



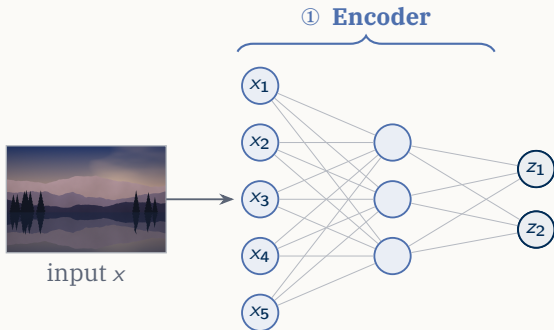
This box is exactly where **autoencoders** come in.

Enter the Autoencoder



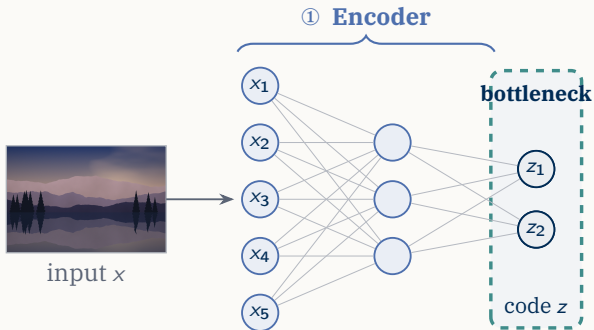
A neural network that learns to **compress** data and **reconstruct** it
It has two main parts: **1. Encoder** **2. Decoder**

Autoencoder Architecture



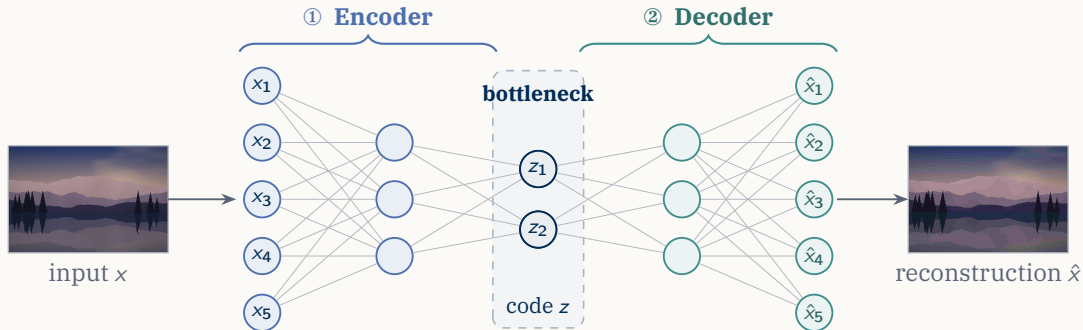
The **encoder** compresses the image, layer by layer, until it reaches the bottleneck.

Autoencoder Architecture



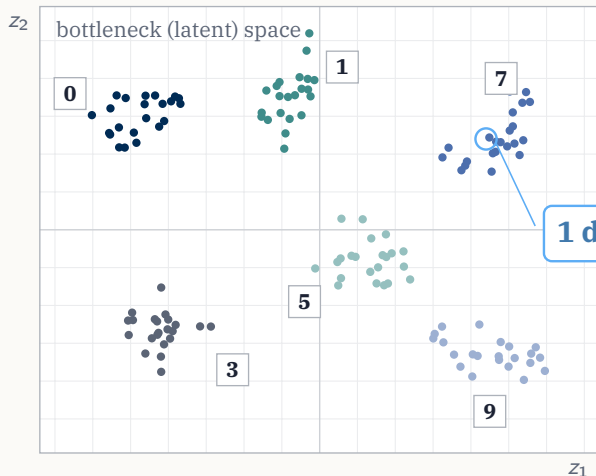
The **bottleneck** allows only the *essential* information to pass which is called *latent code*, represented in the *latent space*

Autoencoder Architecture



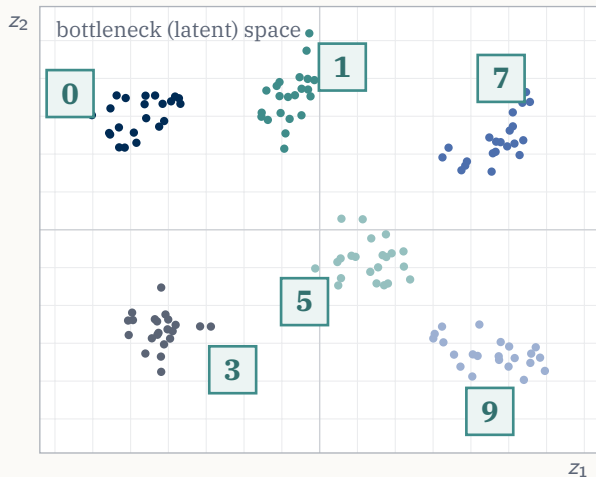
The **decoder** uses that latent code to rebuild the image: $\hat{x} \approx x$.

The Problem with Autoencoders



Here in this bottleneck space, each point is exactly **one image** after compression

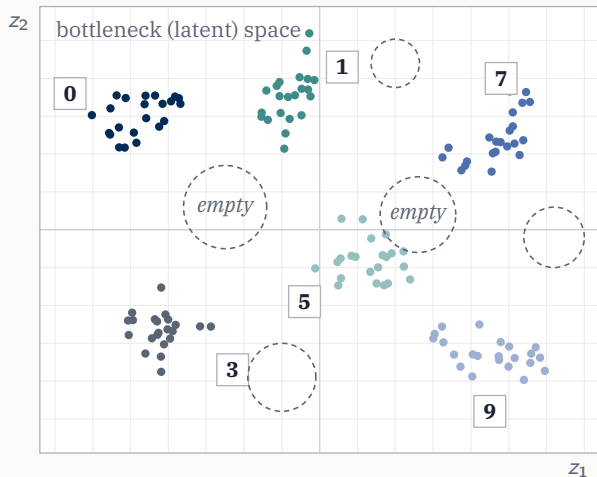
The Problem with Autoencoders



Here in this bottleneck space, each point is exactly **one image** after compression

Notice how similar images form separate **islands**, such as 0, 1, 5, 7, and 9

The Problem with Autoencoders

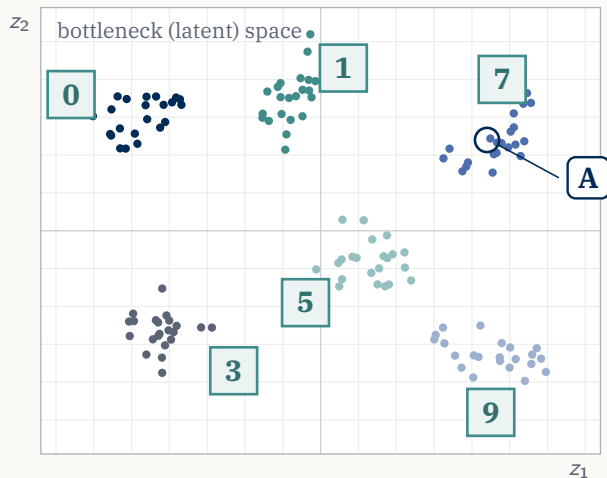


Here in this bottleneck space, each point is exactly **one image** after compression

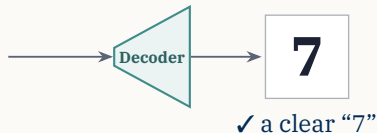
Notice how similar images form separate **islands**, such as 0, 1, 5, 7, and 9

There are lots of **empty spaces** between them. The decoder has not seen training examples from these regions.

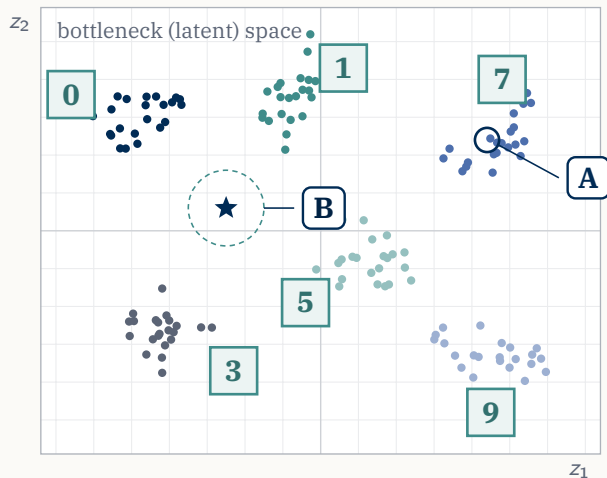
What Lives in the Gaps?



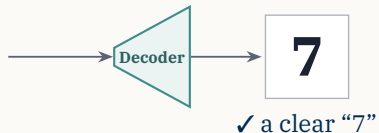
A point inside an island



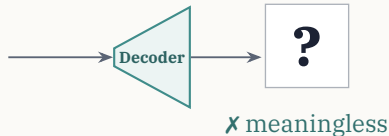
What Lives in the Gaps?



A point inside an island



B random point from an empty gap



The Key Insight

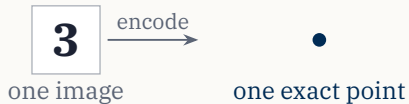
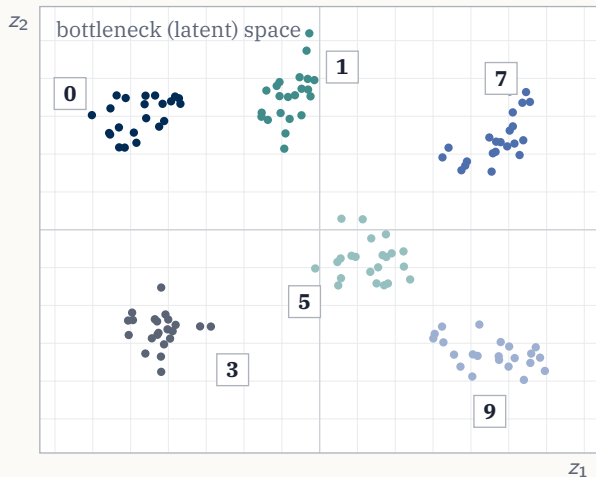
Instead of storing each image as one exact point, what if every image were represented as a **distribution**?

The Key Insight

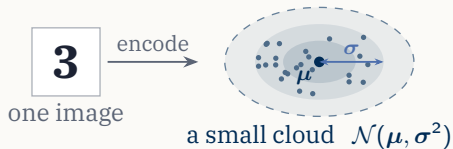
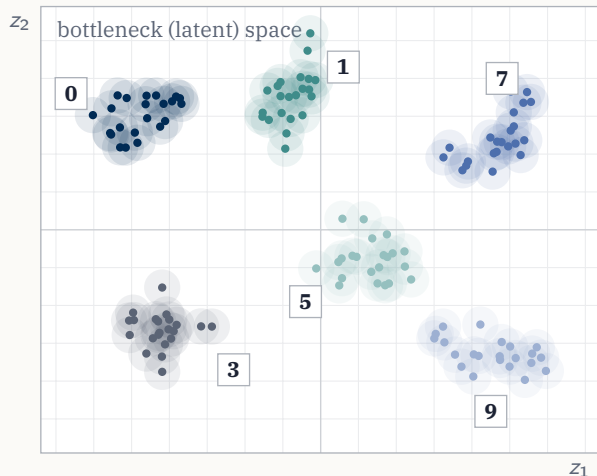
Instead of storing each image as one exact point, what if every image were represented as a **distribution**?

That's the core idea behind the Variational Autoencoder (VAE)

From Points to Distributions



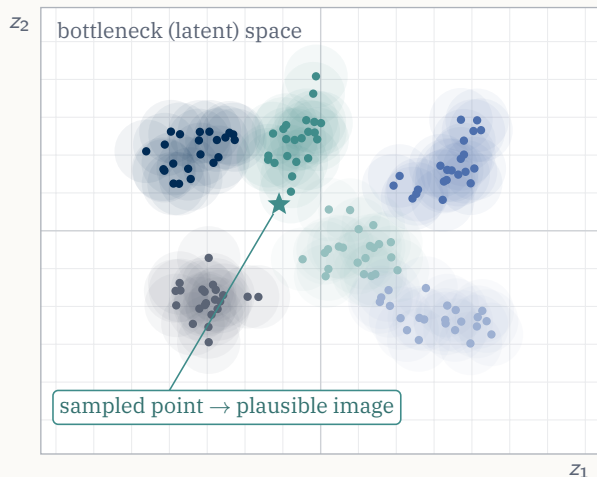
From Points to Distributions



VAE: one image $\rightarrow$ one **distribution**

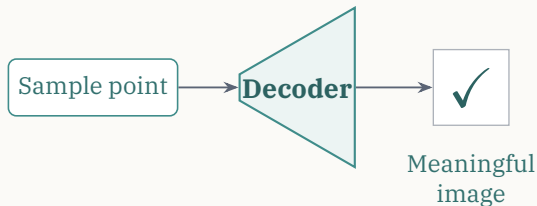
Each image now occupies a **small area**.

From Points to Distributions

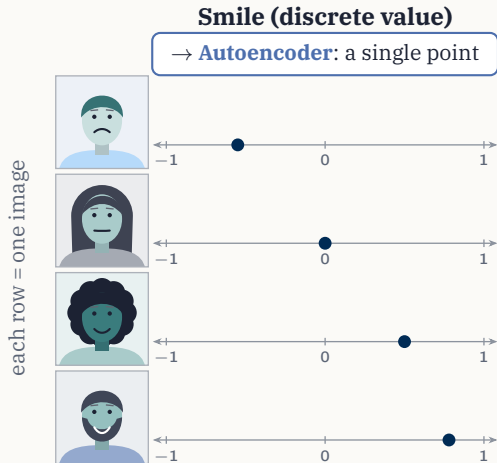


Notice, the distributions overlap, which **reduces empty gaps**.

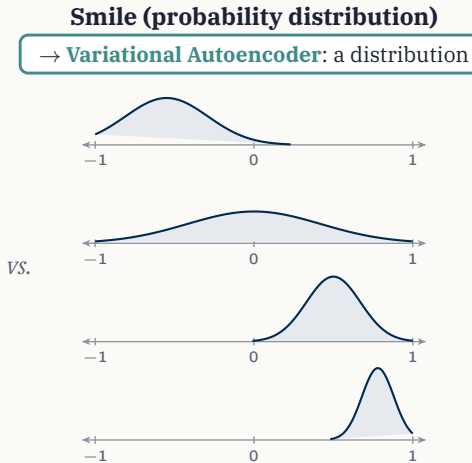
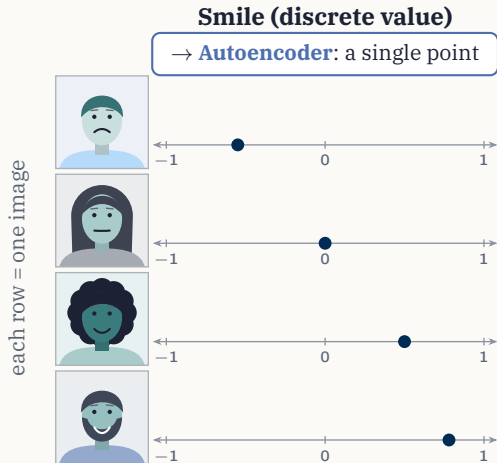
As a result, we can choose a new point from that area and it will still decode into a meaningful image.



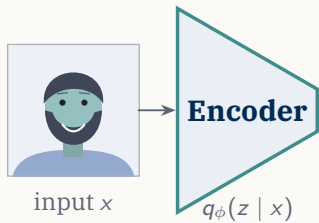
One Image, One Distribution



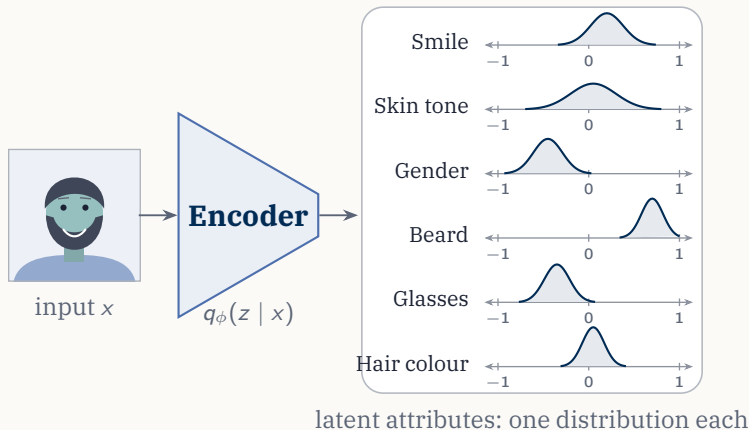
One Image, One Distribution



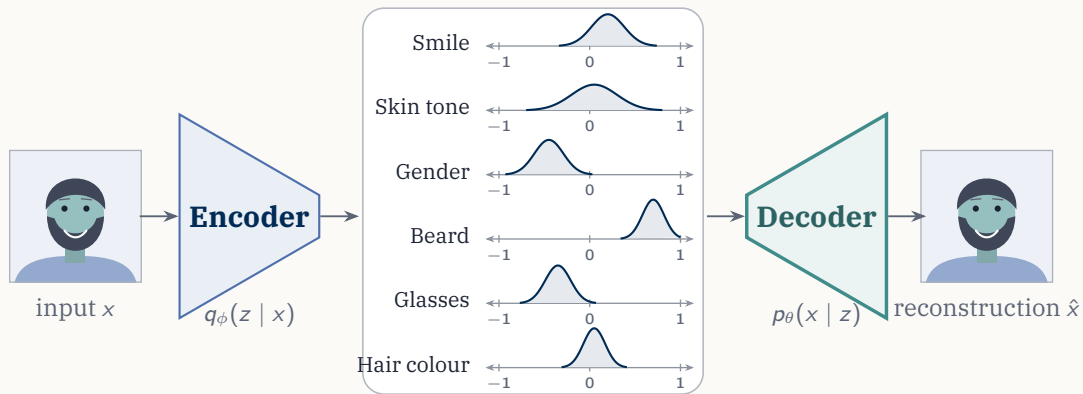
VAE Simplified: Latent Attributes



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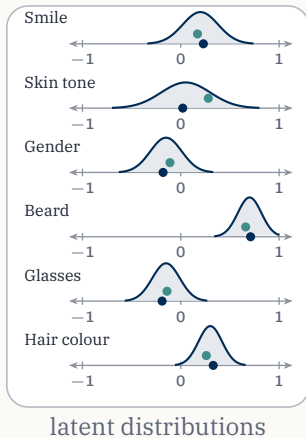


VAE Simplified: Latent Attributes

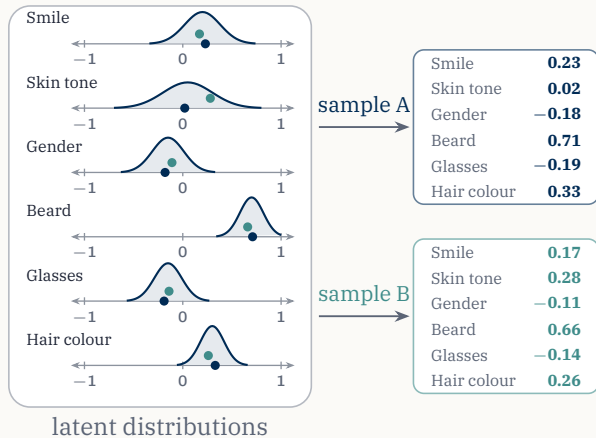


latent attributes: one distribution each

VAE Simplified: Sampling and Decoding



VAE Simplified: Sampling and Decoding



VAE Simplified: Sampling and Decoding

